1) a)

The plane $x = 0$ has normal vector $\mathbf{n} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 2 \\ 3 \\ -9 \end{pmatrix}$

$\cos\theta = \frac{\mathbf{n}\cdot\mathbf{d}}{|\mathbf{n}||\mathbf{d}|} \Rightarrow \cos\theta = \frac{1\times2+0\times3+0\times(-9)}{\sqrt{1^2+0^2+0^2}\sqrt{2^2+3^2+(-9)^2}} = \frac{2}{1\times\sqrt{94}} = \frac{\sqrt{94}}{47}$

$\Rightarrow \theta = \arccos\left(\frac{\sqrt{94}}{47}\right) = 78.095\ldots°$

Angle between line and plane $= 90° - 78.095\ldots° = 11.9°$ (1 DP)

1) b)

The plane $y = 0$ has normal vector $\mathbf{n} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 2 \\ 3 \\ -9 \end{pmatrix}$

$\cos\theta = \frac{\mathbf{n}\cdot\mathbf{d}}{|\mathbf{n}||\mathbf{d}|} \Rightarrow \cos\theta = \frac{0\times2+1\times3+0\times(-9)}{\sqrt{0^2+1^2+0^2}\sqrt{2^2+3^2+(-9)^2}} = \frac{3}{1\times\sqrt{94}} = \frac{3\sqrt{94}}{94}$

$\Rightarrow \theta = \arccos\left(\frac{3\sqrt{94}}{94}\right) = 71.975\ldots°$

Angle between line and plane $= 90° - 71.975\ldots° = 18.0°$ (1 DP)

1) c)

The plane $z = 0$ has normal vector $\mathbf{n} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 2 \\ 3 \\ -9 \end{pmatrix}$

$\cos\theta = \frac{\mathbf{n}\cdot\mathbf{d}}{|\mathbf{n}||\mathbf{d}|} \Rightarrow \cos\theta = \frac{0\times2+0\times3+1\times(-9)}{\sqrt{0^2+0^2+1^2}\sqrt{2^2+3^2+(-9)^2}} = \frac{-9}{1\times\sqrt{94}} = -\frac{9\sqrt{94}}{94}$

$\Rightarrow \theta = \arccos\left(-\frac{9\sqrt{94}}{94}\right) = 158.168\ldots°$

Acute angle between line and normal vector is $180° - 158.168\ldots° = 21.831\ldots°$

Angle between line and plane $= 90° - 21.831\ldots° = 68.2°$ (1 DP)

2)

The plane has normal vector $\mathbf{n} = \begin{pmatrix} 4 \\ 1 \\ -2 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 2 \\ 4 \\ -3 \end{pmatrix}$

$\cos\theta = \frac{\mathbf{n}\cdot\mathbf{d}}{|\mathbf{n}||\mathbf{d}|} \Rightarrow \cos\theta = \frac{4\times2+1\times4+(-2)\times(-3)}{\sqrt{4^2+1^2+(-2)^2}\sqrt{2^2+4^2+(-3)^2}} = \frac{18}{\sqrt{21}\times\sqrt{29}} = \frac{6\sqrt{609}}{203}$

$\Rightarrow \theta = \arccos\left(\frac{6\sqrt{609}}{203}\right) = 43.164\ldots°$

Angle between line and plane $= 90° - 43.164\ldots° = 46.8°$ (1 DP)

3)

The plane has normal vector $\mathbf{n} = \begin{pmatrix} 6 \\ 15 \\ 10 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 5 \\ -8 \\ 9 \end{pmatrix}$

$\cos\theta = \frac{\mathbf{n}\cdot\mathbf{d}}{|\mathbf{n}||\mathbf{d}|} \Rightarrow \cos\theta = \frac{6\times5+15\times(-8)+10\times9}{\sqrt{6^2+15^2+10^2}\sqrt{5^2+(-8)^2+9^2}} = \frac{0}{\sqrt{361}\times\sqrt{170}} = 0$

$\Rightarrow \theta = \arccos(0) = 90°$

So the line is perpendicular to the normal vector of the plane, so the line is parallel to the plane. We must now check whether the line lies in the plane, by checking whether $(3, 7, -3)$ lies in the plane:

$6(3) + 15(7) + 10(-3) - 90 = 18 + 105 - 30 - 90 = 3 \neq 0$

So the line is not contained within the plane, and is just parallel to it.

4)

The plane has normal vector $\mathbf{n} = \begin{pmatrix} 3 \\ 12 \\ -6 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 6 \\ -4 \\ -5 \end{pmatrix}$

$\cos\theta = \frac{\mathbf{n}\cdot\mathbf{d}}{|\mathbf{n}||\mathbf{d}|} \Rightarrow \cos\theta = \frac{3\times6+12\times(-4)+(-6)\times(-5)}{\sqrt{3^2+12^2+(-6)^2}\sqrt{6^2+(-4)^2+(-5)^2}} = \frac{0}{\sqrt{189}\times\sqrt{77}} = 0$

$\Rightarrow \theta = \arccos(0) = 90°$

So the line is perpendicular to the normal vector of the plane, so the line is parallel to the plane. We must now check whether the line lies in the plane, by checking whether $\left(2, \frac{1}{4}, \frac{11}{3}\right)$ lies in the plane:

$3(2) + 12\left(\frac{1}{4}\right) - 6\left(\frac{11}{3}\right) + 13 = 6 + 3 - 22 + 13 = 0$

So the line is contained within the plane.

5)

The plane has normal vector $\mathbf{n} = \begin{pmatrix} 5 \\ 3 \\ 2 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix}$

$\cos\theta = \frac{\mathbf{n}\cdot\mathbf{d}}{|\mathbf{n}||\mathbf{d}|} \Rightarrow \cos\theta = \frac{5\times(-2)+3\times1+2\times4}{\sqrt{5^2+3^2+2^2}\sqrt{(-2)^2+1^2+4^2}} = \frac{1}{\sqrt{38}\times\sqrt{21}} = \frac{\sqrt{798}}{798} \neq 0$

$\Rightarrow \theta = \arccos\left(\frac{\sqrt{798}}{798}\right) = 88.0°$ (1 DP)

So the line intersects the plane at one point.

6) a)

The plane has normal vector $\mathbf{n} = \begin{pmatrix} 1 \\ 1 \\ 10 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 1 \\ 1 \\ -0.5 \end{pmatrix} - \begin{pmatrix} 0.2 \\ 0.4 \\ -0.6 \end{pmatrix} = \begin{pmatrix} 0.8 \\ 0.6 \\ 0.1 \end{pmatrix}$ or $\begin{pmatrix} 8 \\ 6 \\ 1 \end{pmatrix}$

$\cos\theta = \frac{\mathbf{n}\cdot\mathbf{d}}{|\mathbf{n}||\mathbf{d}|} \Rightarrow \cos\theta = \frac{1\times8+1\times6+10\times1}{\sqrt{1^2+1^2+10^2}\sqrt{8^2+6^2+1^2}} = \frac{24}{\sqrt{102}\times\sqrt{101}} = \frac{4\sqrt{10302}}{1717}$

$\Rightarrow \theta = \arccos\left(\frac{4\sqrt{10302}}{1717}\right) = 76.322\ldots°$

Angle between line and plane $= 90° - 76.322\ldots° = 13.7°$ (1 DP)

6) b)

P has the coordinates $(0.8, 92.1, -9.19)$

Let l be the line with equation $\mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ -0.5 \end{pmatrix} + \lambda \begin{pmatrix} 8 \\ 6 \\ 1 \end{pmatrix}$

Let A be a point on the line, so $\overrightarrow{OA} = \begin{pmatrix} 1+8\lambda \\ 1+6\lambda \\ -0.5+\lambda \end{pmatrix}$

$\overrightarrow{PA} = \overrightarrow{OA} - \overrightarrow{OP} = \begin{pmatrix} 1+8\lambda \\ 1+6\lambda \\ -0.5+\lambda \end{pmatrix} - \begin{pmatrix} 0.8 \\ 92.1 \\ -9.19 \end{pmatrix} = \begin{pmatrix} 0.2+8\lambda \\ -91.1+6\lambda \\ 8.69+\lambda \end{pmatrix}$

$\overrightarrow{PA} \cdot \begin{pmatrix} 8 \\ 6 \\ 1 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} 0.2+8\lambda \\ -91.1+6\lambda \\ 8.69+\lambda \end{pmatrix} \cdot \begin{pmatrix} 8 \\ 6 \\ 1 \end{pmatrix} = 0$

$\Rightarrow (0.2+8\lambda) \times 8 + (-91.1+6\lambda) \times 6 + (8.69+\lambda) \times 1 = 0$

$\Rightarrow 1.6 + 6.4\lambda - 546.6 + 36\lambda + 8.69 + \lambda = 0$

$\Rightarrow 101\lambda - 536.31 = 0$

$\lambda = 5.31$

So $\overrightarrow{PA} = \begin{pmatrix} 0.2+8(5.31) \\ -91.1+6(5.31) \\ 8.69+(5.31) \end{pmatrix} = \begin{pmatrix} 42.68 \\ -59.24 \\ 14 \end{pmatrix}$

So $\left|\overrightarrow{PA}\right| = \sqrt{42.68^2 + (-59.24)^2 + 14^2} = 74.3$ m (1 DP)

Alternative Method (using the Vector Product):

Let $\mathbf{p} = \begin{pmatrix} 0.8 \\ 92.1 \\ -9.19 \end{pmatrix}$

Let A be a known point on the line so that its position vector is $\mathbf{a} = \begin{pmatrix} 1 \\ 1 \\ -0.5 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 8 \\ 6 \\ 1 \end{pmatrix}$

Then $h = \frac{|\mathbf{d} \times (\mathbf{p}-\mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix} 8 \\ 6 \\ 1 \end{pmatrix} \times \left(\begin{pmatrix} 0.8 \\ 92.1 \\ -9.19 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ -0.5 \end{pmatrix} \right)\right|}{\left|\begin{pmatrix} 8 \\ 6 \\ 1 \end{pmatrix}\right|} = \frac{\left|\begin{pmatrix} 8 \\ 6 \\ 1 \end{pmatrix} \times \begin{pmatrix} -0.2 \\ 91.1 \\ -8.69 \end{pmatrix}\right|}{\left|\begin{pmatrix} 8 \\ 6 \\ 1 \end{pmatrix}\right|}$

$\begin{pmatrix} 8 \\ 6 \\ 1 \end{pmatrix} \times \begin{pmatrix} -0.2 \\ 91.1 \\ -8.69 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 8 & 6 & 1 \\ -0.2 & 91.1 & -8.69 \end{vmatrix}$

$= (6 \times (-8.69) - 1 \times 91.1)\mathbf{i} - \big(8 \times (-8.69) - 1 \times (-0.2)\big)\mathbf{j} + \big(8 \times 91.1 - 6 \times (-0.2)\big)\mathbf{k}$

$= -143.24\mathbf{i} + 69.32\mathbf{j} + 730\mathbf{k}$

$\therefore h = \frac{\left|\begin{pmatrix} -143.24 \\ 69.32 \\ 730 \end{pmatrix}\right|}{\left|\begin{pmatrix} 8 \\ 6 \\ 1 \end{pmatrix}\right|} = \frac{\sqrt{(-143.24)^2+69.32^2+730^2}}{\sqrt{8^2+6^2+1^2}} = 74.3$ m (1 DP)

7) a)

$\mathbf{r} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 1 \\ -2 \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 1 \\ -2 \end{pmatrix}$

$x = 2 + 5\lambda$

$y = -1 + \lambda$

$z = 3 - 2\lambda$

$x = 0 \Rightarrow 2 + 5\lambda = 0 \Rightarrow \lambda = -\frac{2}{5}$

Substitute into y and z:

$y = -1 + \left(-\frac{2}{5}\right) = -\frac{7}{5}$

$z = 3 - 2\left(-\frac{2}{5}\right) = \frac{19}{5}$

So the line intersects the plane at $\left(0, -\frac{7}{5}, \frac{19}{5}\right)$

7) b)

$y = 0 \Rightarrow -1 + \lambda = 0 \Rightarrow \lambda = 1$

Substitute into x and z:

$x = 2 + 5(1) = 7$

$z = 3 - 2(1) = 1$

So the line intersects the plane at $(7, 0, 1)$

7) c)

$z = 0 \Rightarrow 3 - 2\lambda = 0 \Rightarrow \lambda = \frac{3}{2}$

Substitute into x and y:

$x = 2 + 5\left(\frac{3}{2}\right) = 7$

$y = -1 + \left(\frac{3}{2}\right) = \frac{1}{2}$

So the line intersects the plane at $\left(\frac{19}{2}, \frac{1}{2}, 0\right)$

8)

$$\mathbf{r} = \begin{pmatrix}3\\1\\1\end{pmatrix} + \lambda\begin{pmatrix}0\\2\\5\end{pmatrix} \Rightarrow \begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}3\\1\\1\end{pmatrix} + \lambda\begin{pmatrix}0\\2\\5\end{pmatrix}$$

$x = 3$

$y = 1 + 2\lambda$

$z = 1 + 5\lambda$

$4(3) + (1 + 2\lambda) - (1 + 5\lambda) = 9$

$\Rightarrow 12 + 1 + 2\lambda - 1 - 5\lambda = 9$

$\Rightarrow 3 - 3\lambda = 0$

$\Rightarrow \lambda = 1$

Substitute into y and z:

$y = 1 + 2(1) = 3$

$z = 1 + 5(1) = 6$

So the line intersects the plane at $(3, 3, 6)$

9)

$$\mathbf{r} = \begin{pmatrix}3\\2\\4\end{pmatrix} + \lambda\begin{pmatrix}1\\1\\-1\end{pmatrix} \Rightarrow \begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}3\\2\\4\end{pmatrix} + \lambda\begin{pmatrix}1\\1\\-1\end{pmatrix}$$

$x = 3 + \lambda$

$y = 2 + \lambda$

$z = 4 - \lambda$

$(3 + \lambda) + (2 + \lambda) - (4 - \lambda) = 0$

$\Rightarrow 3 + \lambda + 2 + \lambda - 4 + \lambda = 0$

$\Rightarrow 3\lambda + 1 = 0$

$\Rightarrow \lambda = -\frac{1}{3}$

Substitute into x, y and z:

$x = 3 + \left(-\frac{1}{3}\right) = \frac{8}{3}$

$y = 2 + \left(-\frac{1}{3}\right) = \frac{5}{3}$

$z = 4 - \left(-\frac{1}{3}\right) = \frac{13}{3}$

Let P be the point on the plane closest to A, which has coordinates $\left(\frac{8}{3}, \frac{5}{3}, \frac{13}{3}\right)$

$$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix}\frac{8}{3}\\\frac{5}{3}\\\frac{13}{3}\end{pmatrix} - \begin{pmatrix}3\\2\\4\end{pmatrix} = \begin{pmatrix}-\frac{1}{3}\\-\frac{1}{3}\\\frac{1}{3}\end{pmatrix}$$

Let B be the image of A, reflected in the plane.

$$\overrightarrow{OB} = \overrightarrow{OA} + 2\overrightarrow{AP} = \begin{pmatrix}3\\2\\4\end{pmatrix} + 2\begin{pmatrix}-\frac{1}{3}\\-\frac{1}{3}\\\frac{1}{3}\end{pmatrix} = \begin{pmatrix}\frac{7}{3}\\\frac{4}{3}\\\frac{14}{3}\end{pmatrix}$$

So B has coordinates $\left(\frac{7}{3}, \frac{4}{3}, \frac{14}{3}\right)$

10)

$$\mathbf{r} = \begin{pmatrix} 5 \\ -2 \\ \frac{1}{2} \end{pmatrix} + \lambda \begin{pmatrix} 10 \\ -1 \\ -2 \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5 \\ -2 \\ \frac{1}{2} \end{pmatrix} + \lambda \begin{pmatrix} 10 \\ -1 \\ -2 \end{pmatrix}$$

$x = 5 + 10\lambda$

$y = -2 - \lambda$

$z = \frac{1}{2} - 2\lambda$

$10(5 + 10\lambda) - (-2 - \lambda) - 2\left(\frac{1}{2} - 2\lambda\right) = -54$

$\Rightarrow 50 + 100\lambda + 2 + \lambda - 1 + 4\lambda = -54$

$\Rightarrow 105\lambda + 105 = 0$

$\Rightarrow \lambda = -1$

Substitute into x, y and z:

$x = 5 + 10(-1) = -5$

$y = -2 - (-1) = -1$

$z = \frac{1}{2} - 2(-1) = \frac{5}{2}$

Let P be the point on the plane closest to A, which has coordinates $\left(-5, -1, \frac{5}{2}\right)$

$$\overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA} = \begin{pmatrix} -5 \\ -1 \\ \frac{5}{2} \end{pmatrix} - \begin{pmatrix} 5 \\ -2 \\ \frac{1}{2} \end{pmatrix} = \begin{pmatrix} -10 \\ 1 \\ 2 \end{pmatrix}$$

Let B be the image of A, reflected in the plane.

$$\overrightarrow{OB} = \overrightarrow{OA} + 2\overrightarrow{AP} = \begin{pmatrix} 5 \\ -2 \\ \frac{1}{2} \end{pmatrix} + 2\begin{pmatrix} -10 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -15 \\ 0 \\ \frac{9}{2} \end{pmatrix}$$

So B has coordinates $\left(-15, 0, \frac{9}{2}\right)$

11)

$l_1: \mathbf{r} = \begin{pmatrix}2\\8\\3\end{pmatrix} + \lambda\begin{pmatrix}1\\4\\0\end{pmatrix} \Rightarrow \begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}2\\8\\3\end{pmatrix} + \lambda\begin{pmatrix}1\\4\\0\end{pmatrix}$

$x = 2 + \lambda$

$y = 8 + 4\lambda$

$z = 3$

$9(2 + \lambda) - 20(8 + 4\lambda) + 2(3) = 6$

$\Rightarrow 18 + 9\lambda - 160 - 80\lambda + 6 = 6$

$\Rightarrow -71\lambda - 142 = 0$

$\Rightarrow \lambda = -2$

Substitute into x, y and z:

$x = 2 + (-2) = 0$

$y = 8 + 4(-2) = 0$

$z = 3$

So P has coordinates $(0, 0, 3)$

$l_2: \mathbf{r} = \begin{pmatrix}-1\\0\\0\end{pmatrix} + \mu\begin{pmatrix}2\\0\\1\end{pmatrix} \Rightarrow \begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}-1\\0\\0\end{pmatrix} + \mu\begin{pmatrix}2\\0\\1\end{pmatrix}$

$x = -1 + 2\mu$

$y = 0$

$z = \mu$

$9(-1 + 2\mu) - 20(0) + 2(\mu) = 6$

$\Rightarrow -9 + 18\mu + 0 + 2\mu = 6$

$\Rightarrow 20\mu - 15 = 0$

$\Rightarrow \lambda = \frac{3}{4}$

Substitute into x, y and z:

$x = -1 + 2\left(\frac{3}{4}\right) = \frac{1}{2}$

$y = 0$

$z = \frac{3}{4}$

So Q has coordinates $\left(\frac{1}{2}, 0, \frac{3}{4}\right)$

$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = \begin{pmatrix}\frac{1}{2}\\0\\\frac{3}{4}\end{pmatrix} - \begin{pmatrix}0\\0\\3\end{pmatrix} = \begin{pmatrix}\frac{1}{2}\\0\\-\frac{9}{4}\end{pmatrix}$

$|\overrightarrow{PQ}| = \sqrt{\left(\frac{1}{2}\right)^2 + 0^2 + \left(-\frac{9}{4}\right)^2} = \sqrt{\frac{85}{16}} = \frac{\sqrt{85}}{4}$ units

12) a)

If the line l lies in the plane, then $\begin{pmatrix} b \\ 4 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = 0$

$\Rightarrow b \times 2 + 4 \times 1 + 1 \times 1 = 0$

$\Rightarrow 2b + 5 = 0$

$\Rightarrow b = -\frac{5}{2}$

And $(2, a, 3)$ must lie on the plane:

$2(2) + (a) + (3) = 5$

$\Rightarrow a + 7 = 5$

$\Rightarrow a = -2$

$\therefore a = -2$ and $b = -\frac{5}{2}$

12) b)

a can be any real number, and $b \neq -\frac{5}{2}$

13) a)

$-3 = 0 - \lambda \Rightarrow \lambda = 3$

$7 = 1 + 2\lambda \Rightarrow \lambda = 3$

$5 = 2 + \lambda \Rightarrow \lambda = 3$

$\therefore P$ lies on the line l

13) b)

$-x + 2y + z = d$

Substitute in the point $P(-3, 7, 5)$:

$-(-3) + 2(7) + (5) = d \Rightarrow d = 22$

$\therefore \Pi: -x + 2y + z = 22$

13) c)

R lies on the line and so its coordinates can be written as $(-\lambda, 1+2\lambda, 2+\lambda)$

So R has position vector

$$\overrightarrow{OR} = \begin{pmatrix} -\lambda \\ 1+2\lambda \\ 2+\lambda \end{pmatrix}$$

$$\overrightarrow{QR} = \overrightarrow{OR} - \overrightarrow{OQ} = \begin{pmatrix} -\lambda \\ 1+2\lambda \\ 2+\lambda \end{pmatrix} - \begin{pmatrix} -3 \\ 7-\frac{1}{\sqrt{5}} \\ 5+\frac{2}{\sqrt{5}} \end{pmatrix} = \begin{pmatrix} -\lambda+3 \\ 2\lambda+\frac{-30+\sqrt{5}}{5} \\ \lambda-\frac{15+2\sqrt{5}}{5} \end{pmatrix}$$

$$\left|\overrightarrow{QR}\right| = \sqrt{2} \Rightarrow (-\lambda+3)^2 + \left(2\lambda+\frac{-30+\sqrt{5}}{5}\right)^2 + \left(\lambda-\frac{15+2\sqrt{5}}{5}\right)^2 = 25$$

$$\Rightarrow \lambda^2 - 6\lambda + 9 + 4\lambda^2 + \frac{-120+4\sqrt{5}}{5}\lambda + \frac{181-12\sqrt{5}}{5} + \lambda^2 - \frac{30+4\sqrt{5}}{5}\lambda + \frac{49+12\sqrt{5}}{5} = 25$$

$$\Rightarrow 6\lambda^2 - 36\lambda + 30 = 0$$

$$\Rightarrow \lambda^2 - 6\lambda + 5 = 0$$

$$\Rightarrow (\lambda-5)(\lambda-1) = 0$$

$\Rightarrow \lambda = 1$ or $\lambda = 5$

When $\lambda = 1$, R has coordinates $(-1, 3, 3)$

When $\lambda = 5$, R has coordinates $(-5, 11, 7)$

14) a)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 4 \\ -1 \\ -2 \end{pmatrix} - \begin{pmatrix} 3 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ -5 \end{pmatrix}$$

$$l_1: \mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -5 \end{pmatrix}$$

14) b)

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} - \begin{pmatrix} 3 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} -2 \\ 4 \\ -2 \end{pmatrix}$$

$$l_2: \mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 4 \\ -2 \end{pmatrix} \text{ or } l_2: \mathbf{r} = \begin{pmatrix} 3 \\ -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

14) c)

$$\begin{pmatrix}1\\1\\-5\end{pmatrix}\cdot\begin{pmatrix}3\\2\\1\end{pmatrix}=1\times3+1\times2+(-5)\times1=3+2-5=0$$

$$\begin{pmatrix}1\\-2\\1\end{pmatrix}\cdot\begin{pmatrix}3\\2\\1\end{pmatrix}=1\times3+(-2)\times2+1\times1=3-4+1=0$$

$\therefore \mathbf{n}$ is perpendicular to both l_1 and l_2

14) d)

Π has the equation $3x+2y+z+d=0$

Substitute $A(3,-2,3)$ to find d:

$3(3)+2(-2)+(3)+d=0 \Rightarrow d=-8$

$\therefore \Pi: 3x+2y+z-8=0$

14) e)

$$\overrightarrow{DE}=\overrightarrow{OE}-\overrightarrow{OD}=\begin{pmatrix}6\\4\\10\end{pmatrix}-\begin{pmatrix}8\\5\\9\end{pmatrix}=\begin{pmatrix}-2\\-1\\1\end{pmatrix}$$

$$l_3: \mathbf{r}=\begin{pmatrix}8\\5\\9\end{pmatrix}+\lambda\begin{pmatrix}-2\\-1\\1\end{pmatrix}$$

Substitute into Π:

$3(8-2\lambda)+2(5-\lambda)+(9+\lambda)-8=0$

$\Rightarrow 24-6\lambda+10-2\lambda+9+\lambda-8=0$

$\Rightarrow -7\lambda+35=0$

$\Rightarrow \lambda=5$

Substitute $\lambda=5$ into l_3:

$P\big(8-2(5),5-(5),9+(5)\big)=P(-2,0,14)$

14) f)

The plane has normal vector $\mathbf{n} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$

$\cos\theta = \frac{\mathbf{n}\cdot\mathbf{d}}{|\mathbf{n}||\mathbf{d}|} \Rightarrow \cos\theta = \frac{3\times(-2)+2\times(-1)+1\times1}{\sqrt{3^2+2^2+1^2}\sqrt{(-2)^2+(-1)^2+1^2}} = -\frac{7}{\sqrt{14}\times\sqrt{6}} = -\frac{\sqrt{21}}{6}$

$\Rightarrow \theta = \arccos\left(-\frac{\sqrt{21}}{6}\right) = 139.797\ldots°$

Angle between line and plane $= 90° - (180° - 139.797\ldots°) = 49.8°$ (1 DP)

14) g)

Find a unit vector in the direction of $\mathbf{d} = \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$:

$\hat{\mathbf{d}} = \frac{1}{\sqrt{(-2)^2+(-1)^2+1^2}}\begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{6}}\begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$

$\overrightarrow{OF} = \overrightarrow{OP} \pm \mathbf{d} = \begin{pmatrix} -2 \\ 0 \\ 14 \end{pmatrix} \pm \frac{1}{\sqrt{6}}\begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$

$= \begin{pmatrix} -2-\frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ 14+\frac{1}{\sqrt{6}} \end{pmatrix} = \begin{pmatrix} \frac{-6-\sqrt{6}}{3} \\ -\frac{\sqrt{6}}{6} \\ \frac{84+\sqrt{6}}{6} \end{pmatrix}$ or $\begin{pmatrix} -2+\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ 14-\frac{1}{\sqrt{6}} \end{pmatrix} = \begin{pmatrix} \frac{-6+\sqrt{6}}{3} \\ \frac{\sqrt{6}}{6} \\ \frac{84-\sqrt{6}}{6} \end{pmatrix}$

So $F\left(\frac{-6-\sqrt{6}}{3}, -\frac{\sqrt{6}}{6}, \frac{84+\sqrt{6}}{6}\right)$ or $F\left(\frac{-6+\sqrt{6}}{3}, \frac{\sqrt{6}}{6}, \frac{84-\sqrt{6}}{6}\right)$

15) a)

$$\begin{pmatrix}3\\-9\\2\end{pmatrix}\cdot\begin{pmatrix}1\\4\\3\end{pmatrix}=3\times1+(-9)\times4+2\times3=-27\neq0$$

So the line is not parallel to the plane and cannot be contained within it. Hence the line must intersect the plane at one point.

$$\mathbf{r}\cdot\begin{pmatrix}3\\-9\\2\end{pmatrix}=18\Rightarrow3x-9y+2z=18$$

$$3(2+\lambda)-9(2+4\lambda)+2(5+3\lambda)=18$$
$$\Rightarrow6+3\lambda-18-36\lambda+10+6\lambda=18$$
$$\Rightarrow-27\lambda-20=0$$
$$\Rightarrow\lambda=-\frac{20}{27}$$

So the point where the line intersects the plane has position vector

$$\begin{pmatrix}2+\left(-\frac{20}{27}\right)\\2+4\left(-\frac{20}{27}\right)\\5+3\left(-\frac{20}{27}\right)\end{pmatrix}=\begin{pmatrix}\frac{34}{27}\\-\frac{26}{27}\\\frac{25}{9}\end{pmatrix}$$

So the line intersects the plane at $\left(\frac{34}{27},-\frac{26}{27},\frac{25}{9}\right)$

15) b)

$$\begin{pmatrix}3\\-9\\2\end{pmatrix}\cdot\begin{pmatrix}7\\13\\48\end{pmatrix}=3\times7+(-9)\times13+2\times48=0$$

So the line is parallel to the plane. Now check if $(6,-3,2)$ lies in the plane:

$3(6)-9(-3)+2(2)=18+27+4=49\neq18$

So the point does not lie in the plane. So the line is parallel to the plane but does not lie in the plane.

15) c)

$\frac{x-6}{2} = \frac{y-4}{4} = \frac{z-18}{15} \Rightarrow \mathbf{r} = \begin{pmatrix} 6 \\ 4 \\ 18 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 4 \\ 15 \end{pmatrix}$

$\begin{pmatrix} 3 \\ -9 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 4 \\ 15 \end{pmatrix} = 3 \times 2 + (-9) \times 4 + 2 \times 15 = 0$

So the line is parallel to the plane. Now check if $(6, 4, 18)$ lies in the plane:

$3(6) - 9(4) + 2(18) = 18 - 36 + 36 = 18$

So the point does lie on the plane. So the line lies in the plane.

15) d)

$x + 8 = 3 - y = \frac{z+1}{6} \Rightarrow \frac{x+8}{1} = \frac{y-3}{-1} = \frac{z+1}{6} \Rightarrow \mathbf{r} = \begin{pmatrix} -8 \\ 3 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 6 \end{pmatrix}$

$\begin{pmatrix} 3 \\ -9 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \\ 6 \end{pmatrix} = 3 \times 1 + (-9) \times (-1) + 2 \times 6 = 24 \neq 0$

So the line is not parallel to the plane and cannot be contained within it. Hence the line must intersect the plane at one point.

$3(-8 + \lambda) - 9(3 - \lambda) + 2(-1 + 6\lambda) = 18$

$\Rightarrow -24 + 3\lambda - 27 + 9\lambda - 2 + 12\lambda = 18$

$\Rightarrow 24\lambda - 71 = 0$

$\Rightarrow \lambda = \frac{71}{24}$

So the point where the line intersects the plane has position vector

$\begin{pmatrix} -8 + \left(\frac{71}{24}\right) \\ 3 - \left(\frac{71}{24}\right) \\ -1 + 6\left(\frac{71}{24}\right) \end{pmatrix} = \begin{pmatrix} -\frac{121}{24} \\ \frac{1}{24} \\ \frac{67}{4} \end{pmatrix}$

So the line intersects the plane at $\left(-\frac{121}{24}, \frac{1}{24}, \frac{67}{4}\right)$

16) a)

$l_1: \frac{x+5}{3} = y-1 = 2-z \Rightarrow \frac{x+5}{3} = \frac{y-1}{1} = \frac{z-2}{-1} \Rightarrow \mathbf{r} = \begin{pmatrix} -5 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}$

$l_2: 5-x = y+3 = \frac{z-5}{2} \Rightarrow \frac{x-5}{-1} = \frac{y+3}{1} = \frac{z-5}{2} \Rightarrow \mathbf{r} = \begin{pmatrix} 5 \\ -3 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$

$-5 + 3\lambda = 5 - \mu \Rightarrow 3\lambda + \mu = 10$ (1)

$1 + \lambda = -3 + \mu \Rightarrow \lambda - \mu = -4$ (2)

$2 - \lambda = 5 + 2\mu \Rightarrow -\lambda - 2\mu = 3$ (3)

Solve (1) and (2): $\lambda = \frac{3}{2}$ and $\mu = \frac{11}{2}$

Check in (3):

$-\left(\frac{3}{2}\right) - 2\left(\frac{11}{2}\right) = \frac{19}{2} \neq 3$

So the lines do not intersect.

l_1 and l_2 are skew lines

16) b)

$$\begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix} \times \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 1 & -1 \\ -1 & 1 & 2 \end{vmatrix}$$

$= (1 \times 2 - (-1) \times 1)\mathbf{i} - \big(3 \times 2 - (-1) \times (-1)\big)\mathbf{j} + \big(3 \times 1 - 1 \times (-1)\big)\mathbf{k}$

$= 3\mathbf{i} - 5\mathbf{j} + 4\mathbf{k}$

So $\begin{pmatrix} 3 \\ -5 \\ 4 \end{pmatrix}$ is a vector that is perpendicular to both l_1 and l_2

16) c)

$3x - 5y + 4z = d$ is parallel to both l_1 and l_2

Substitute in $(5, -3, 5)$ to find the plane that also contains l_2:

$3(5) - 5(-3) + 4(5) = d \Rightarrow d = 50$

$\therefore \Pi_1: 3x - 5y + 4z = 50$

16) d)

$3x - 5y + 4z = d$ is parallel to both l_1 and l_2

Substitute in $(-5, 1, 2)$ to find the plane that also contains l_1:

$3(-5) - 5(1) + 4(2) = d \Rightarrow d = -12$

$\therefore \Pi_2: 3x - 5y + 4z = -12$

16) e)

Substitute l_3 into Π_1:

$3(2 - 3\alpha) - 5(5 + 3\alpha) + 4(6 + 8\alpha) = 50$

$\Rightarrow 6 - 9\alpha - 25 - 15\alpha + 24 + 32\alpha = 50$

$\Rightarrow 8\alpha + 5 = 50$

$\Rightarrow \alpha = \frac{45}{8}$

So the position vector of the point of intersection between l_3 and Π_1 is

$$\overrightarrow{OP} = \begin{pmatrix} 2 - 3\left(\frac{45}{8}\right) \\ 5 + 3\left(\frac{45}{8}\right) \\ 6 + 8\left(\frac{45}{8}\right) \end{pmatrix} = \begin{pmatrix} -\frac{119}{8} \\ \frac{175}{8} \\ 51 \end{pmatrix}$$

Substitute l_3 into Π_2:

$3(2 - 3\alpha) - 5(5 + 3\alpha) + 4(6 + 8\alpha) = -12$

$\Rightarrow 6 - 9\alpha - 25 - 15\alpha + 24 + 32\alpha = -12$

$\Rightarrow 8\alpha + 5 = -12$

$\Rightarrow \alpha = -\frac{17}{8}$

So the position vector of the point of intersection between l_3 and Π_1 is

$$\overrightarrow{OQ} = \begin{pmatrix} 2 - 3\left(-\frac{17}{8}\right) \\ 5 + 3\left(-\frac{17}{8}\right) \\ 6 + 8\left(-\frac{17}{8}\right) \end{pmatrix} = \begin{pmatrix} \frac{67}{8} \\ -\frac{11}{8} \\ -11 \end{pmatrix}$$

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = \begin{pmatrix} \frac{67}{8} \\ -\frac{11}{8} \\ -11 \end{pmatrix} - \begin{pmatrix} -\frac{119}{8} \\ \frac{175}{8} \\ 51 \end{pmatrix} = \begin{pmatrix} \frac{93}{4} \\ -\frac{93}{4} \\ -62 \end{pmatrix}$$

$$\left|\overrightarrow{PQ}\right| = \sqrt{\left(\frac{93}{4}\right)^2 + \left(-\frac{93}{4}\right)^2 + (-62)^2} = \sqrt{\frac{39401}{8}} = \frac{31\sqrt{82}}{4} \text{ units}$$

17)

$2x+8=\frac{4-y}{3}, z=2 \Rightarrow \frac{x+4}{\frac{1}{2}}=\frac{y-4}{-3}, z=2 \Rightarrow \mathbf{r}=\begin{pmatrix}-4\\4\\2\end{pmatrix}+\alpha\begin{pmatrix}\frac{1}{2}\\-3\\0\end{pmatrix}$

Find the cartesian equation of the plane:

$\begin{pmatrix}2\\-1\\2\end{pmatrix}\times\begin{pmatrix}7\\3\\5\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&-1&2\\7&3&5\end{vmatrix}$

$=\big((-1)\times 5-2\times 3\big)\mathbf{i}-(2\times 5-2\times 7)\mathbf{j}+(2\times 3-(-1)\times 7)\mathbf{k}$

$=-11\mathbf{i}+4\mathbf{j}+13\mathbf{k}$

So the equation of the plane is $-11x+4y+13z+d=0$

Substitute in $(1,1,4)$ to find d:

$-11(1)+4(1)+13(4)+d=0 \Rightarrow 45+d=0 \Rightarrow d=-45$

$\therefore -11x+4y+13z-45=0$

Substitute the line into the plane equation:

$-11\left(-4+\frac{1}{2}\alpha\right)+4(4-3\alpha)+13(2)-45=0$

$\Rightarrow 44-\frac{11}{2}\alpha+16-12\alpha+26-45=0$

$\Rightarrow \frac{35}{2}\alpha-41=0$

$\Rightarrow \alpha=\frac{82}{35}$

So the point where the line intersects the plane has position vector

$\begin{pmatrix}-4+\frac{1}{2}\left(\frac{82}{35}\right)\\4-3\left(\frac{82}{35}\right)\\2\end{pmatrix}=\begin{pmatrix}-\frac{99}{35}\\-\frac{106}{35}\\2\end{pmatrix}$

So the line intersects the plane at $\left(-\frac{99}{35},-\frac{106}{35},2\right)$

18) a)

$\mathbf{b}=\begin{pmatrix}50\\120\\235\end{pmatrix}-\begin{pmatrix}60\\70\\320\end{pmatrix}=\begin{pmatrix}-10\\50\\-85\end{pmatrix}$

$\mathbf{c}=\begin{pmatrix}130\\100\\155\end{pmatrix}-\begin{pmatrix}50\\120\\235\end{pmatrix}=\begin{pmatrix}80\\-20\\-80\end{pmatrix}$

$\therefore \mathbf{r}=\begin{pmatrix}60\\70\\320\end{pmatrix}+\lambda\begin{pmatrix}-10\\50\\-85\end{pmatrix}+\mu\begin{pmatrix}80\\-20\\-80\end{pmatrix}$

18) b)

$$\begin{pmatrix}3\\4\\2\end{pmatrix}\cdot\begin{pmatrix}-10\\50\\-85\end{pmatrix}=3\times(-10)+4\times 50+2\times(-85)=0$$

$$\begin{pmatrix}3\\4\\2\end{pmatrix}\cdot\begin{pmatrix}80\\-20\\-80\end{pmatrix}=3\times 80+4\times(-20)+2\times(-80)=0$$

So the vector $3\mathbf{i}+4\mathbf{j}+2\mathbf{k}$ is perpendicular to the plane

18) c)

The plane has equation $3x+4y+2z=d$

Substitute $(60,70,320)$ into the equation to find d:

$3(60)+4(70)+2(320)=d \Rightarrow d=1100$

$\therefore 3x+4y+2z=1100$

18) d)

The line through $(20, 65, 100)$ that is perpendicular to the plane has equation:

$$\mathbf{r} = \begin{pmatrix} 20 \\ 65 \\ 100 \end{pmatrix} + \alpha \begin{pmatrix} 3 \\ 4 \\ 2 \end{pmatrix}$$

Substitute into the plane to find the point of intersection:

$3(20 + 3\alpha) + 4(65 + 4\alpha) + 2(100 + 2\alpha) = 1100$

$\Rightarrow 60 + 9\alpha + 260 + 16\alpha + 200 + 4\alpha = 1100$

$\Rightarrow 29\alpha - 580 = 0$

$\Rightarrow \alpha = 20$

The point of intersection has position vector $\begin{pmatrix} 20 + 3(20) \\ 50 + 4(20) \\ 100 + 2(20) \end{pmatrix} = \begin{pmatrix} 80 \\ 130 \\ 140 \end{pmatrix}$

The tunnel should begin at $(80, 130, 140)$

The length of the tunnel is $\sqrt{(80 - 20)^2 + (130 - 50)^2 + (140 - 100)^2}$

$= \sqrt{11600} = 20\sqrt{29}$ m

Alternative Method:

$$\mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} 20 \\ 65 \\ 100 \end{pmatrix}$$

$d = -1100$

$$\frac{|n_1x_1 + n_2y_1 + n_3z_1 + d|}{\sqrt{n_1^2 + n_2^2 + n_3^2}} = \frac{|3 \times 20 + 4 \times 65 + 2 \times 100 - 1100|}{\sqrt{3^2 + 4^2 + 2^2}} = \frac{580}{\sqrt{29}} = 20\sqrt{29} \text{ m}$$

19) a)

$$\begin{pmatrix} 4 \\ -1 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix} = 4 \times 3 + (-1) \times 2 + (-1) \times 2 = 12 - 2 - 2 = 8 \neq 0$$

So Π_1 and Π_2 are not perpendicular

19) b)

Let $z = 0$:

$4x - y = 3$ (1)

$3x + 2y = 9$ (2)

Solve (1) and (2) to get $x = \frac{15}{11}$ and $y = \frac{27}{11}$

So $\left(\frac{15}{11}, \frac{27}{11}, 0\right)$ is a point that lies on both Π_1 and Π_2

19) c)

$$\begin{pmatrix} 4 \\ -1 \\ -1 \end{pmatrix} \times \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & -1 & -1 \\ 3 & 2 & 2 \end{vmatrix}$$

$$= \big((-1) \times 2 - (-1) \times 2\big)\mathbf{i} - (4 \times 2 - (-1) \times 3)\mathbf{j} + (4 \times 2 - (-1) \times 3)\mathbf{k}$$

$$= 0\mathbf{i} - 11\mathbf{j} + 11\mathbf{k}$$

So $\begin{pmatrix} 0 \\ -11 \\ 11 \end{pmatrix}$ or $\begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$ is parallel to both Π_1 and Π_2

19) d)

Π_3 has the equation $-y + z = d$

Substitute in the point $(5, 3, 1)$ to find d:

$-3 + 1 = d \Rightarrow d = -2$

$\therefore \Pi_3 : -y + z = -2$

20) a)

Determine where the line l intersects the plane Π:

$5(4+2\lambda)+8(3+6\lambda)+9(-5+3\lambda)=16$

$\Rightarrow 20+10\lambda+24+48\lambda-45+27\lambda=16$

$\Rightarrow 85\lambda-17=0$

$\Rightarrow \lambda=\frac{1}{5}$

So the line intersects the plane at $\left(4+2\left(\frac{1}{5}\right),3+6\left(\frac{1}{5}\right),-5+3\left(\frac{1}{5}\right)\right)=\left(\frac{22}{5},\frac{21}{5},-\frac{22}{5}\right)$

Find a vector of length 14 in the direction of the line l:

$\frac{14}{\sqrt{2^2+6^2+3^2}}\begin{pmatrix}2\\6\\3\end{pmatrix}=\frac{14}{7}\begin{pmatrix}2\\6\\3\end{pmatrix}=\begin{pmatrix}4\\12\\6\end{pmatrix}$

So the points P and Q have position vectors $\begin{pmatrix}\frac{22}{5}\\\frac{21}{5}\\-\frac{22}{5}\end{pmatrix}\pm\begin{pmatrix}4\\12\\6\end{pmatrix}=\begin{pmatrix}\frac{42}{5}\\\frac{81}{5}\\\frac{8}{5}\end{pmatrix}$ or $\begin{pmatrix}\frac{2}{5}\\-\frac{39}{5}\\-\frac{52}{5}\end{pmatrix}$

So P and Q have coordinates $\left(\frac{42}{5},\frac{81}{5},\frac{8}{5}\right)$ and $\left(\frac{2}{5},-\frac{39}{5},-\frac{52}{5}\right)$

20) b)

A is the point with coordinates $(11, -6, 1)$

Let B be a point on the line, so $\overrightarrow{OB} = \begin{pmatrix} 4+2\lambda \\ 3+6\lambda \\ -5+3\lambda \end{pmatrix}$

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 4+2\lambda \\ 3+6\lambda \\ -5+3\lambda \end{pmatrix} - \begin{pmatrix} 11 \\ -6 \\ 1 \end{pmatrix} = \begin{pmatrix} -7+2\lambda \\ 9+6\lambda \\ -6+3\lambda \end{pmatrix}$

$\overrightarrow{AB} \cdot \begin{pmatrix} 2 \\ 6 \\ 3 \end{pmatrix} = 0 \Rightarrow \begin{pmatrix} -7+2\lambda \\ 9+6\lambda \\ -6+3\lambda \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 6 \\ 3 \end{pmatrix} = 0$

$\Rightarrow (-7+2\lambda) \times 2 + (9+6\lambda) \times 6 + (-6+3\lambda) \times 3 = 0$

$\Rightarrow -14 + 4\lambda + 54 + 36\lambda - 18 + 9\lambda = 0$

$\Rightarrow 49\lambda + 22 = 0$

$\lambda = -\frac{22}{49}$

So $\overrightarrow{AP} = \begin{pmatrix} -7+2\left(-\frac{22}{49}\right) \\ 9+6\left(-\frac{22}{49}\right) \\ -6+3\left(-\frac{22}{49}\right) \end{pmatrix} = \begin{pmatrix} -\frac{387}{49} \\ \frac{309}{49} \\ -\frac{360}{49} \end{pmatrix}$

So $\left|\overrightarrow{AP}\right| = \sqrt{\left(-\frac{387}{49}\right)^2 + \left(\frac{309}{49}\right)^2 + \left(-\frac{360}{49}\right)^2} = \sqrt{\frac{7650}{49}} = \frac{15\sqrt{34}}{7}$

So the area of the triangle APQ is $\frac{1}{2} \times 28 \times \frac{15\sqrt{34}}{7} = 30\sqrt{34}$ units^2

Alternative Method (using the Vector Product):

Let $\mathbf{p} = \begin{pmatrix} 11 \\ -6 \\ 1 \end{pmatrix}$

Let A be a known point on the line so that its position vector is $\mathbf{a} = \begin{pmatrix} 4 \\ 3 \\ -5 \end{pmatrix}$

Let $\mathbf{d} = \begin{pmatrix} 2 \\ 6 \\ 3 \end{pmatrix}$

Then $h = \frac{|\mathbf{d} \times (\mathbf{p} - \mathbf{a})|}{|\mathbf{d}|} = \frac{\left|\begin{pmatrix} 2 \\ 6 \\ 3 \end{pmatrix} \times \left(\begin{pmatrix} 11 \\ -6 \\ 1 \end{pmatrix} - \begin{pmatrix} 4 \\ 3 \\ -5 \end{pmatrix}\right)\right|}{\left|\begin{pmatrix} 2 \\ 6 \\ 3 \end{pmatrix}\right|} = \frac{\left|\begin{pmatrix} 2 \\ 6 \\ 3 \end{pmatrix} \times \begin{pmatrix} 7 \\ -9 \\ 6 \end{pmatrix}\right|}{\left|\begin{pmatrix} 2 \\ 6 \\ 3 \end{pmatrix}\right|}$

$\begin{pmatrix} 2 \\ 6 \\ 3 \end{pmatrix} \times \begin{pmatrix} 7 \\ -9 \\ 6 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 6 & 3 \\ 7 & -9 & 6 \end{vmatrix}$

$= \left(6 \times 6 - 3 \times (-9)\right)\mathbf{i} - (2 \times 6 - 3 \times 7)\mathbf{j} + (2 \times (-9) - 6 \times 7)\mathbf{k}$

$= 63\mathbf{i} + 9\mathbf{j} - 60\mathbf{k}$

$\therefore h = \frac{\left|\begin{pmatrix} 63 \\ 9 \\ -60 \end{pmatrix}\right|}{\left|\begin{pmatrix} 2 \\ 6 \\ 3 \end{pmatrix}\right|} = \frac{\sqrt{63^2+9^2+(-60)^2}}{\sqrt{2^2+6^2+3^2}} = \frac{\sqrt{7650}}{\sqrt{49}} = \frac{15\sqrt{34}}{7}$

So the area of the triangle APQ is $\frac{1}{2} \times 28 \times \frac{15\sqrt{34}}{7} = 30\sqrt{34}$ units^2

21)

Firstly, find where the line intersects the plane:

$(2+\lambda)-(3-7\lambda)+(-8+6\lambda)=6$

$\Rightarrow 14\lambda-15=0$

$\Rightarrow \lambda=\frac{15}{14}$

So the line intersects the plane at $\left(2+\left(\frac{15}{14}\right),3-7\left(\frac{15}{14}\right),-8+6\left(\frac{15}{14}\right)\right)=\left(\frac{43}{14},-\frac{9}{2},-\frac{11}{7}\right)$

Now we need to reflect the point $(2,3,-8)$ in the plane. To do this, first find the closest point on the plane to $(2,3,-8)$:

The line that goes through $(2,3,-8)$ and is perpendicular to the plane is

$\mathbf{r}=\begin{pmatrix}2\\3\\-8\end{pmatrix}+\mu\begin{pmatrix}1\\-1\\1\end{pmatrix}$

Substitute the line into the plane equation:

$(2+\mu)-(3-\mu)+(-8+\mu)=6$

$\Rightarrow 3\mu-15=0$

$\Rightarrow \mu=5$

So the line intersects the plane at the point

$\big(2+(5),3-(5),-8+(5)\big)=(7,-2,-3)$

Find the vector from $(2,3,-8)$ to $(7,-2,-3)$: $\begin{pmatrix}7\\-2\\-3\end{pmatrix}-\begin{pmatrix}2\\3\\-8\end{pmatrix}=\begin{pmatrix}5\\-5\\5\end{pmatrix}$

So the image of the point $(2,3,-8)$ reflected in the plane has position vector:

$\begin{pmatrix}2\\3\\-8\end{pmatrix}+2\begin{pmatrix}5\\-5\\5\end{pmatrix}=\begin{pmatrix}12\\-7\\2\end{pmatrix}$

So the reflected line passes through $(12,-7,2)$ and $\left(\frac{43}{14},-\frac{9}{2},-\frac{11}{7}\right)$

$\begin{pmatrix}\frac{43}{14}\\-\frac{9}{2}\\-\frac{11}{7}\end{pmatrix}-\begin{pmatrix}12\\-7\\2\end{pmatrix}=\begin{pmatrix}-\frac{125}{14}\\\frac{5}{2}\\-\frac{25}{7}\end{pmatrix}$

So the equation of the line is $\mathbf{r}=\begin{pmatrix}12\\-7\\2\end{pmatrix}+\alpha\begin{pmatrix}-\frac{125}{14}\\\frac{5}{2}\\-\frac{25}{7}\end{pmatrix}$ or $\mathbf{r}=\begin{pmatrix}12\\-7\\2\end{pmatrix}+\alpha\begin{pmatrix}25\\-7\\10\end{pmatrix}$

22)

$\begin{pmatrix}16\\-50\\23\end{pmatrix}\cdot\begin{pmatrix}6\\1\\-2\end{pmatrix}=16\times6+(-50)\times1+23\times(-2)=0$

$\therefore l$ is perpendicular to the plane Π_1

Substitute $(22,-64,30)$ into Π_1:

$6(22)+(-64)-2(30)=8$

As $(22,-64,30)$ lies in Π_1, then l lies in Π_1

$\begin{pmatrix}16\\-50\\23\end{pmatrix}\cdot\begin{pmatrix}5\\-3\\-10\end{pmatrix}=16\times5+(-50)\times(-3)+23\times(-10)=0$

$\therefore l$ is perpendicular to the plane Π_2

Substitute $(22,-64,30)$ into Π_2:

$5(22)-3(-64)-10(30)=2$

As $(22,-64,30)$ lies in Π_2, then l lies in Π_2

23)

$\begin{pmatrix}2\\-1\\1\end{pmatrix}\times\begin{pmatrix}3\\1\\-1\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\2&-1&1\\3&1&-1\end{vmatrix}$

$=\big((-1)\times(-1)-1\times1\big)\mathbf{i}-(2\times(-1)-1\times3)\mathbf{j}+(2\times1-(-1)\times3)\mathbf{k}$

$=0\mathbf{i}+5\mathbf{j}+5\mathbf{k}$

Find a point common to both planes:

Let $y=1$, then:

$2x-1+z=5\Rightarrow2x+z=6$ (1)

$3x+1-z=2\Rightarrow3x-z=1$ (2)

Solve (1) and (2): $x=\frac{7}{5}$ and $z=\frac{16}{5}$

$\therefore\mathbf{r}=\begin{pmatrix}\frac{7}{5}\\1\\\frac{16}{5}\end{pmatrix}+\lambda\begin{pmatrix}0\\5\\5\end{pmatrix}$ or $\mathbf{r}=\begin{pmatrix}\frac{7}{5}\\1\\\frac{16}{5}\end{pmatrix}+\lambda\begin{pmatrix}0\\1\\1\end{pmatrix}$

24)

$$\begin{pmatrix}0\\0\\1\end{pmatrix}\times\begin{pmatrix}5\\1\\3\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\0&0&1\\5&1&3\end{vmatrix}$$

$$=(0\times3-1\times1)\mathbf{i}-(0\times3-1\times5)\mathbf{j}+(0\times1-0\times5)\mathbf{k}$$

$$=-\mathbf{i}+5\mathbf{j}+0\mathbf{k}$$

Find a point common to both planes:

Let $x=1$, then:

$z=0$ (1)

$5+y+3z=8\Rightarrow y+3z=3$ (2)

Solve (1) and (2): $y=3$ and $z=0$

$$\therefore \mathbf{r}=\begin{pmatrix}1\\3\\0\end{pmatrix}+\lambda\begin{pmatrix}-1\\5\\0\end{pmatrix}$$

25) a)

$$\begin{pmatrix}1\\0\\0\end{pmatrix}\times\begin{pmatrix}1\\0\\-4\end{pmatrix}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&0&0\\1&0&-4\end{vmatrix}$$

$$=(0\times(-4)-0\times0)\mathbf{i}-(1\times(-4)-0\times1)\mathbf{j}+(1\times0-0\times1)\mathbf{k}$$

$$=0\mathbf{i}+4\mathbf{j}+0\mathbf{k}$$

Find a point common to both planes:

$x=0$ (1)

$x-4z+20=0\Rightarrow x-4z=-20$ (2)

Solve (1) and (2): $x=0$ and $z=5$

Let $y=1$

$$\therefore \mathbf{r}=\begin{pmatrix}0\\1\\5\end{pmatrix}+\lambda\begin{pmatrix}0\\4\\0\end{pmatrix}\text{ or }\mathbf{r}=\begin{pmatrix}0\\1\\5\end{pmatrix}+\lambda\begin{pmatrix}0\\1\\0\end{pmatrix}$$

25) b)

$$\cos\theta=\frac{\begin{pmatrix}1\\0\\0\end{pmatrix}\cdot\begin{pmatrix}1\\0\\-4\end{pmatrix}}{\sqrt{1^2+0^2+0^2}\times\sqrt{1^2+0^2+(-4)^2}}$$

$$\Rightarrow\cos\theta=\frac{1\times1+0\times0+0\times(-4)}{\sqrt{1}\times\sqrt{17}}$$

$$\Rightarrow\cos\theta=\frac{1}{\sqrt{17}}$$

$\Rightarrow\theta=75.963\ldots°=76.0°$ (3 SF)